

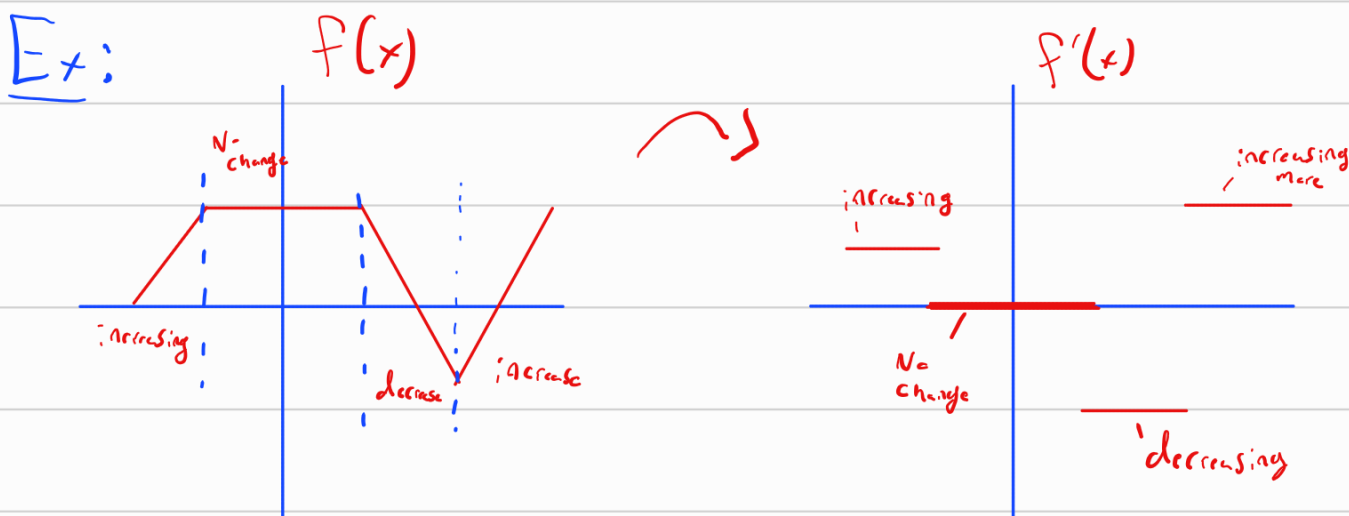
Thus far we've considered the derivative only at a point a . However, we can more generally consider it as a function whose input is x in the domain of f and output is the slope of the graph of f .

The derivative of f is

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

and $f'(x)$ is a function of x as well.

Always remember, $f'(x)$ is just a function that tells slope of f at the point x .



We can also use the limit definition to find the derivative explicitly.

Ex: $f(x) = x^3 - x$

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\overbrace{(x+h)^3 - (x+h)}^{f(x+h)} - \overbrace{(x^3 - x)}^{f(x)}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{x} + 3x^2h + 3xh^2 + h^3 - \cancel{x} - h - \cancel{x^3} + \cancel{x}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{3x^2\cancel{h} + 3xh^2 + h^3 - \cancel{h}}{\cancel{h}}$$

$$= \lim_{h \rightarrow 0} 3x^2 + \underbrace{3xh}_{h=0} + \underbrace{h^2}_{h=0} - 1$$

$f'(x) = 3x^2 - 1$

Ex:

$$f(x) = \frac{1-x}{2+x}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\frac{1-(x+h)}{2+x+h} - \frac{1-x}{2+x}}{h} = \lim_{h \rightarrow 0} \frac{(1-x-h)(2+x) - (2+x+h)(1-x)}{h(2+x+h)(2+x)}$$
$$= \lim_{h \rightarrow 0} \frac{\cancel{2} - \cancel{2x} - 2h + \cancel{x} \cdot \cancel{x^2} - \cancel{xh} - \cancel{2} - \cancel{x} - h + \cancel{2x} + \cancel{x^2} + \cancel{hx}}{h(2+x+h)(2+x)}$$

$$= \lim_{h \rightarrow 0} \frac{-2h - h}{h(2+x+h)(2+x)}$$

$$= \lim_{h \rightarrow 0} \frac{-3}{(2+x+h)(2+x)}$$

$$= \lim_{h \rightarrow 0} - \frac{3}{\underset{h=0}{(2+x+h)}(2+x)}$$

$$f'(x) = - \frac{3}{(2+x)^2}$$

Notation:

Sometimes we use other notation for the derivative of $y=f(x)$

$$f'(x) = \underline{y'} = \frac{dy}{dx} = \underline{\frac{df}{dx}} = \frac{d}{dx} f(x) = Df(x) = D_x f(x)$$

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 most common

Note: The derivative won't always exist.

Ex: $f(x) = |x|$

$$|x| = \begin{cases} x & x > 0 \\ -x & x < 0 \\ 0 & x = 0 \end{cases}$$

if $x > 0$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{x+h-x}{h} = \lim_{h \rightarrow 0} \frac{h}{h} = 1$$

$x < 0$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{-(x+h) - (-x)}{h} = \lim_{h \rightarrow 0} \frac{-x-h+x}{h} = \lim_{h \rightarrow 0} \frac{-h}{h} = -1$$

but at $x=0$, $f'(0)$ DNE as the limit from left and right differ.

Thm:

If f is differentiable at a , then f is continuous at a .

Pf/ If f is differentiable, $f'(a)$ exists

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$\lim_{x \rightarrow a} (x-a) f'(a) = \lim_{x \rightarrow a} f(x) - f(a)$$

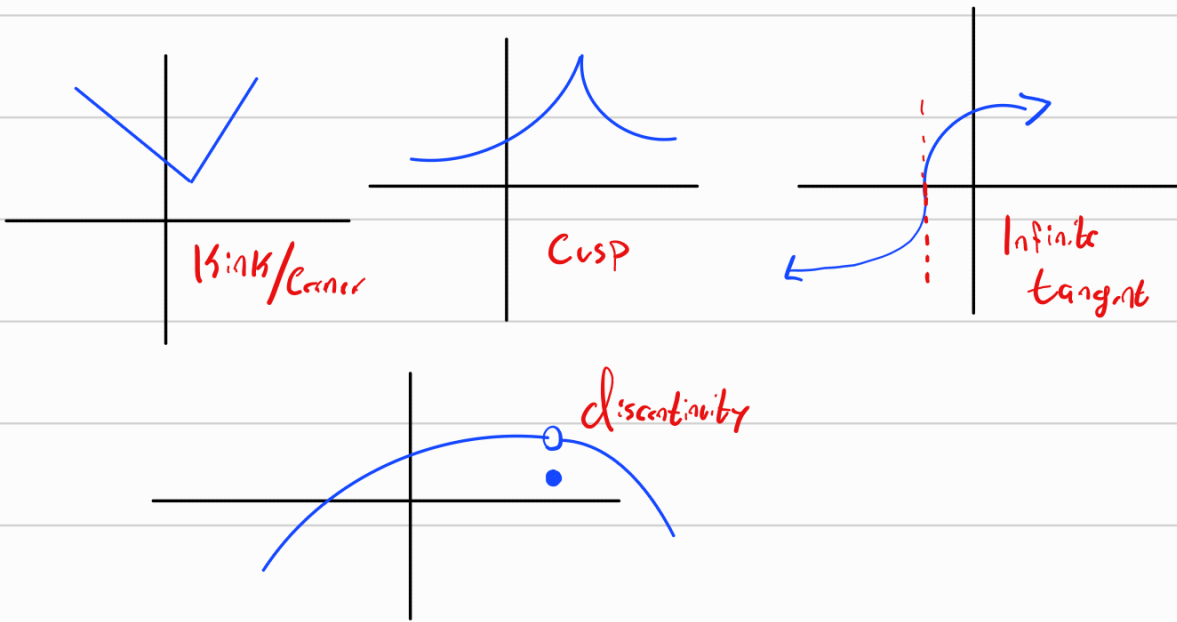
$$0 = \lim_{x \rightarrow a} f(x) - f(a)$$

$$f(a) = \lim_{x \rightarrow a} f(x)$$

How a function fails to be differentiable

There are three cases.

1. Kinks/corners or cusps
2. Infinite discontinuity
3. Discontinuous function



Higher derivatives

We have just been working with the first derivatives of functions, but we can continue taking derivatives if we want

$$\begin{array}{ccc} \frac{dy}{dx}, & \frac{d^2y}{dx^2}, & \frac{d^3y}{dx^3} \\ f'(x), & f''(x), & f'''(x) \\ \text{first} & \text{second} & \text{third derivative} \end{array}$$

Ex: $f(x) = x^3 - x$

We saw earlier,

$$f'(x) = 3x^2 - 1$$

$$f''(x) = \lim_{h \rightarrow 0} \frac{f'(x+h) - f'(x)}{h} = \lim_{h \rightarrow 0} \frac{3(x+h)^2 - 1 - (3x^2 - 1)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{3x^2} + 6xh + \cancel{h^2} - \cancel{1} - \cancel{3x^2} + \cancel{1}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{6xh + h^2}{h}$$

$$= \lim_{h \rightarrow 0} 6x$$

$$f''(x) = 6x$$

$$f'''(x) = \lim_{h \rightarrow 0} \frac{f''(x+h) - f''(x)}{h} = \lim_{h \rightarrow 0} \frac{6(x+h) - 6x}{h} = \lim_{h \rightarrow 0} \frac{\cancel{6x} + 6h - \cancel{6x}}{h}$$
$$= \lim_{h \rightarrow 0} \frac{6h}{h} = 6$$

$$f'''(x) = 6$$

$$f^{(4)}(x) = \lim_{h \rightarrow 0} \frac{f'''(x+h) - f'''(x)}{h} = \lim_{h \rightarrow 0} \frac{6 - 6}{h} = 0$$

$$f^{(4)}(x) = 0$$